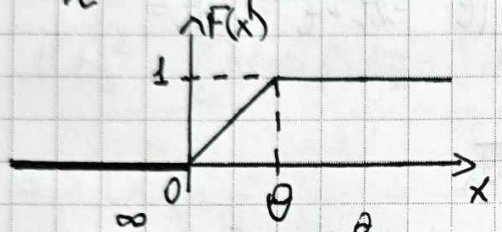
Математическая статистика

Первое Задание

1. $\xi \sim R[0,\theta]$

$\tilde{\theta}_1 = 2\bar{x}$; $\tilde{\theta}_2 = x_{min}$; $\tilde{\theta}_3 = x_{max}$; $\tilde{\theta}_4 = \left(x_1 + \frac{\sum_{k=2}^{n} x_k}{(n-1)}\right)$

$\bar{x}_n$ - выборка объема n



$$p(x) = \frac{1}{\theta}\{0,\theta\}_\theta$$

$$M\xi = \int_{-\infty}^{\infty} x\,p(x)\,dx = \int_0^\theta \frac{x}{\theta}\,dx = \frac{\theta}{2}$$

$$M\xi^2 = \int_{-\infty}^{\infty} x^2 p(x)\,dx = \int_0^\theta \frac{x^2}{\theta}\,dx = \frac{\theta^2}{3}$$

$$D\xi = \frac{\theta^2}{3} - \frac{\theta^2}{4} = \frac{\theta^2}{12}$$

• $\tilde{\theta}_1 = 2\bar{x}$

$$M\tilde{\theta}_1 = M\left[\frac{2}{n}\sum_{i=1}^{n} x_i\right] = \frac{2}{n}\sum_{i=1}^{n} M x_i = 2M\xi = \theta$$

$\Rightarrow$ $\tilde{\theta}_1$ - несмещенная

$$D\tilde{\theta}_1 = D\left[\frac{2}{n}\sum_{i=1}^{n} x_i\right] = \frac{4}{n^2}\sum_{i=1}^{n} D x_i = \frac{4}{n} D\xi = \frac{\theta^2}{3n} \xrightarrow[n\to\infty]{} 0$$

$\Rightarrow$ Состоятельная по дост усл

• $\tilde{\theta}_2 = x_{min}$

$$\Phi(y) = 1 - (1 - F(y))^n$$

$$\varphi(y) = \Phi'(y) = n(1 - F(y))^{n-1} \cdot p(y)$$

$$M\tilde{\theta}_2 = \int_{-\infty}^{\infty} y\varphi(y)\,dy = \int_0^\theta n\left(1 - \frac{y}{\theta}\right)^{n-1} \frac{1}{\theta} y\,dy = \left\{\begin{array}{l} t = 1 - \frac{y}{\theta} \\ dt = -dy/\theta \end{array}\right\} =$$

$$= -\int_1^0 n t^{n-1}(1-t)\theta\,dt = \int_0^1 n\theta t^{n-1}\,dt - \int_0^1 n\theta t^n\,dt = \theta\left(1 - \frac{n}{n+1}\right) = \frac{\theta}{n+1}$$

$\Rightarrow \tilde{\Theta}_2$ – смещенная

$\tilde{\Theta}_2' = (n+1)\tilde{\Theta}_2$ ~~[redacted]~~ ~~не смещ~~ несмещ $M\tilde{\Theta}_2' = \Theta$

$$M\tilde{\Theta}_2^2 = \int_0^\theta n\left(1-\frac{y}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}y^2dy = \left\{ t = 1-\frac{y}{\theta} \right\} =$$

$$= -\int_1^0 nt^{n-1}\theta^2(1-t)^2dt = n\theta^2\int_0^1\left(t^{n-1}-2t^n+t^{n+1}\right)dt =$$

$$= n\theta^2\left(\frac{1}{n}-\frac{2}{n+1}+\frac{1}{n+2}\right) = \frac{2\theta^2}{(n+1)(n+2)}$$

$$D\tilde{\Theta}_2 = \frac{2\theta^2}{(n+1)(n+2)} - \frac{\theta^2}{(n+1)^2} = \frac{2(n+1)-(n+2)}{(n+1)^2(n+2)}\theta^2 =$$

$$= \frac{n}{(n+1)^2(n+2)}\theta^2 \xrightarrow[n\to\infty]{} 0$$

$$D\tilde{\Theta}_2' = \frac{n}{n+2}\theta^2 \not\xrightarrow[n\to\infty]{} 0$$

$\tilde{\Theta}_2'$ по определению

$$\forall\theta>0\ \forall\varepsilon>0\ P(|\tilde{\Theta}_2'-\theta|\geq\varepsilon) \xrightarrow[n\to\infty]{} 0$$

$$P(|\tilde{\Theta}_2'-\theta|\geq\varepsilon) \geq P(\Theta_2'\geq\theta+\varepsilon) = P((n+1)x_{min}\geq\theta+\varepsilon) =$$

$$= P\left(x_{min}\geq\frac{\theta+\varepsilon}{n+1}\right) = 1-P\left(x_{min}<\frac{\theta+\varepsilon}{n+1}\right) = 1-\left(1-F\left(\frac{\theta+\varepsilon}{n+1}\right)\right)^n =$$

$$= \left(1-\left(\frac{\theta+\varepsilon}{\theta(n+1)}\right)\right)^n \xrightarrow[n\to\infty]{} e^{-\frac{+\varepsilon}{\theta}} > 0$$ не состоятельная

- $\tilde{\Theta}_3 = x_{max}$:

$$M\tilde{\Theta}_3 = \int_{-\infty}^{\infty} z\varphi(z)dz \Theta$$

~~[redacted]~~ $\Phi(z) = F^n(z)$ $\varphi(z) = \Phi'(z) = nF^{n-1}(z)\cdot p(z) =$

$$= n\left(\frac{z}{\theta}\right)^{n-1}\cdot\frac{1}{\theta}$$

$$\textcircled{=} \int_0^{\theta} n \cdot \frac{z^n}{\theta^n} dz = \frac{n}{n+1}\theta \Rightarrow \text{смещенная}$$

$$\tilde{\theta}_3' = \frac{n+1}{n}\tilde{\theta}_3 \quad \text{не смещ.} \quad M\tilde{\theta}_3' = \theta$$

$$D\tilde{\theta}_3' = \left(\frac{n+1}{n}\right)^2 D\tilde{\theta}_3$$

$$M\tilde{\theta}_3^2 = \int_{-\infty}^{\infty} z^2 \varphi(z) dz = \int_0^{\theta} n \frac{z^{n+1}}{\theta^n} dz = \frac{n}{n+2}\theta^2$$

$$\Rightarrow D\tilde{\theta}_3 = \frac{n \cdot \theta^2}{(n+2)(n+1)^2}$$

$$D\tilde{\theta}_3' = \frac{(n+1)^2}{n^2} \cdot \frac{n \cdot \theta^2}{(n+2)(n+1)^2} \xrightarrow[n\to\infty]{} 0 \Rightarrow \tilde{\theta}_3' \text{ состоятельна}$$

Исследуем состоятельность $\tilde{\theta}_3$:

$$P(|\tilde{\theta}_3 - \theta| \geq \varepsilon) = P(x_{max} \leq \theta - \varepsilon) + P(x_{max} \geq \theta + \varepsilon) =$$

$$= F^n(\theta - \varepsilon) = \begin{cases} \left(\frac{\theta-\varepsilon}{\theta}\right)^n \xrightarrow[n\to\infty]{} 0, & \varepsilon < \theta \\ 0^n \xrightarrow[n\to\infty]{} 0, & \varepsilon \geq \theta \end{cases}$$

$\Rightarrow \tilde{\theta}_3$ - состоятельна

- $\tilde{\theta}_4 = \left(\frac{x_1 + \sum_{k=2}^{n} x_k}{(n-1)}\right)$

$$M\tilde{\theta}_4 = \frac{Mx_1 + \sum_{k=2}^{n} Mx_k}{n-1} = \frac{\theta}{2} + (n-1)\frac{\theta}{2} \cdot \frac{1}{n-1} = \theta$$

$\Rightarrow$ несмещ.

$$D\tilde{\theta}_4 = Dx_1 + \sum_{k=2}^{n} Dx_k \cdot \frac{1}{(n-1)^2} = \left(1 + \frac{1}{n-1}\right)\frac{\theta^2}{12} = \frac{n-1+1}{n-1}\frac{\theta^2}{12} \not\xrightarrow[n\to\infty]{} 0$$

$\tilde{\theta}_4$ по определению:

$$\tilde{\theta}_4 = \underbrace{x_1}_{\xi_n} + \underbrace{\frac{1}{n-1}\sum_{i=2}^{n} x_i}_{\eta_n}$$

Применим ЗБЧ Хинчина: $\frac{1}{n}\sum_{i=1}^{n}\xi_i \xrightarrow{P} M\xi$

$x_1 \xrightarrow{P} x_1$

$$\frac{1}{n-1}\sum_{i=2}^{n} x_i \xrightarrow{P} M\xi = \frac{\theta}{2}$$

$$\tilde{\theta}_4 = \tilde{x}_1 + \frac{1}{n-1}\sum_{i=2}^{n} x_i \xrightarrow{P} x_1 + \frac{\theta}{2} \to \theta \Rightarrow$$ несостоятельна

θ_1	θ_3	θ_3'
несмещен. состоят	смещ сост	несмещ сост

$$D\theta_1 = \frac{\theta^2}{3n} \qquad D\tilde{\theta}_3' = \frac{\theta^2}{n(n+2)}$$

$$\forall\theta>0 \quad \frac{\theta^2}{n(n+2)} < \frac{\theta^2}{3n} \Rightarrow \tilde{\theta}_3'$$ самая эффективная

Исследуем $\tilde{\theta}_3'$ по определению на состоят.:

$$P(|\tilde{\theta}_3' - \theta| \geq \varepsilon) \xrightarrow[n\to\infty]{?} 0$$

$$P(|\tilde{\theta}_3' - \theta| \geq \varepsilon) = P\left(\max_{i=1\ldots n} x_i \cdot \frac{n+1}{n} \leq \theta - \varepsilon\right) +$$

$$+ P\left(\max_{i=1\ldots n} x_i \cdot \frac{n+1}{n} \geq \theta + \varepsilon\right) = P\left(\max_{i=1\ldots n} x_i \leq \frac{n(\theta-\varepsilon)}{n+1}\right) +$$

$$+ P\left(\max_{i=1\ldots n} x_i \geq \frac{n(\theta+\varepsilon)}{n+1}\right) = \underbrace{F^n\left(\frac{n(\theta-\varepsilon)}{n+1}\right)}_{F_1} + 1 - \underbrace{F^n\left(\frac{n(\theta+\varepsilon)}{n+1}\right)}_{F_2} =$$

$$= F_1 + 1 - F_2$$

F_1: $\Theta > \varepsilon$: $\left(\frac{n(\Theta-\varepsilon)}{\Theta(n+1)}\right)^n = \frac{(\Theta-\varepsilon)}{\Theta(n+\frac{1}{n})} \xrightarrow[n\to\infty]{} 0$

$\Theta < \varepsilon$: $(0)^n \xrightarrow[n\to\infty]{} 0$

F_2: $\frac{n(\Theta+\varepsilon)}{n+1} > \Theta$: $F_2 = 1^n \xrightarrow[n\to\infty]{} 1$

$\frac{n(\Theta+\varepsilon)}{n+1} \leq \Theta$: $n\Theta + \varepsilon n \leq \Theta n + \Theta$

т.е $n \leq \frac{\Theta}{\varepsilon}$

$\Rightarrow \exists N: \forall n \geq N \hookrightarrow \frac{n(\Theta+\varepsilon)}{n+1} > \Theta$

$\Rightarrow F_2 \xrightarrow[n\to\infty]{} 1$

$P(|\tilde{\Theta}_3' - \Theta| \geq \varepsilon) = 1 - F_1 + F_2 \to 1 - 1 + 0 \to 0$

$\Rightarrow$ по опр $\tilde{\Theta}_3'$ состоятельна